

Exp 5	Design of High Pass Filter using Windowing Techniques
Date:	5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Code:

```

clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)

fs = 8000; % Sampling frequency (Hz)

N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- 1. FIR High Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'high', rectwin(N+1));

disp('--- High-Pass Coefficients (Rectangular Window) ---');

disp(b_rect);

%% --- 2. FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm);

%% --- 3. FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');

disp(b_hann);

```

%% --- Frequency Response Plots ---

figure;

freqz(b_rect,1);

title('High-pass FIR using Rectangular Window');

figure;

freqz(b_hamm,1);

title('High-pass FIR using Hamming Window');

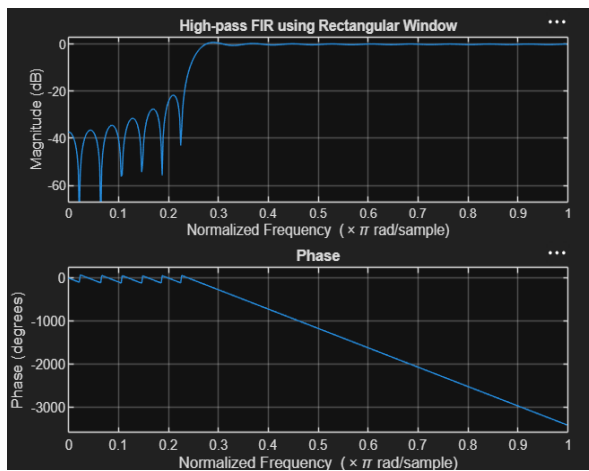
figure;

freqz(b_hann,1);

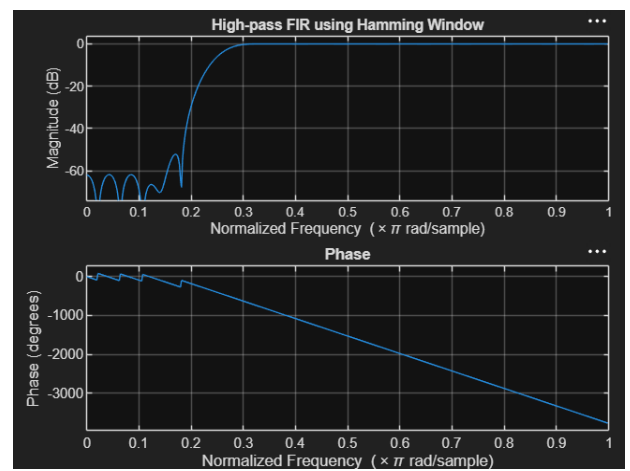
title('High-pass FIR using Hanning Window');

Output:

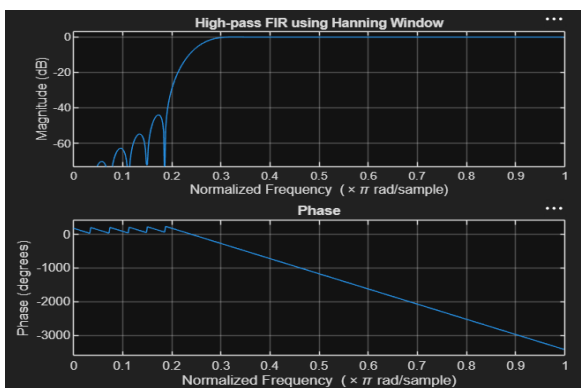
Objective 1:



Objective 2:



Objective 3:



Exp 5	Design of High Pass Filter using Windowing Techniques
Date:	5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

Code:

```

clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)

fs = 8000; % Sampling frequency (Hz)

N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');

disp(b_hamm);

%% --- Frequency Response Plot ---

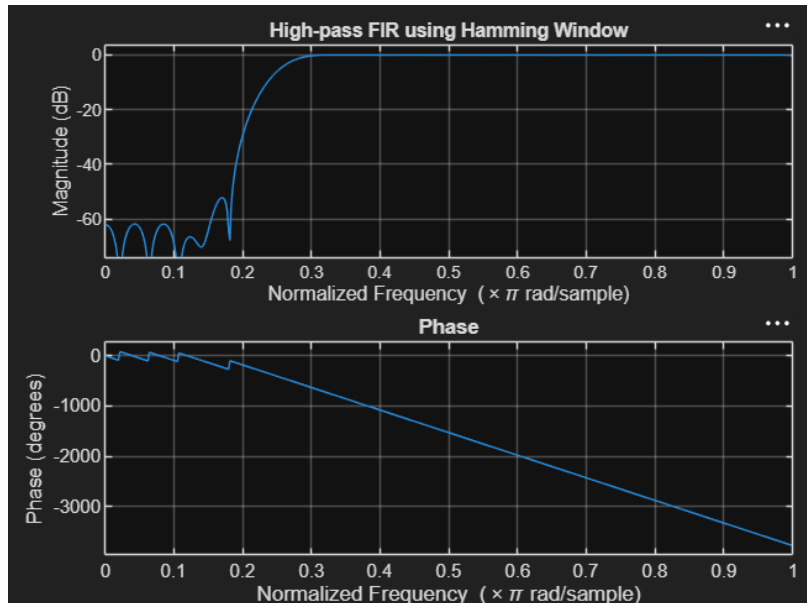
figure;

freqz(b_hamm,1);

title('High-pass FIR using Hamming Window');
```

Output:

Objective:



Exp 5	Design of High Pass Filter using Windowing Techniques
Date:	5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

Code:

```
clc;
clear;
close all;

%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hanning Window ---
```

```

b_hann = fir1(N, wc, 'high', hanning(N+1));

disp('--- High-Pass Coefficients (Hanning Window) ---');

disp(b_hann);

%% --- Frequency Response Plot ---

figure;

freqz(b_hann,1);

title('High-pass FIR using Hanning Window');

```

Output:

Objective:

